

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element F Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to assess whether your proposed solution is viable and realistic, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit, be sure to delete the instructions.

Element F: Consideration of Design Viability

Judgment about Design Viability (F1)

"Design viability" is the capacity of the proposed solution to solve the problem. Design viability refers to how your design decisions are tethered to your design requirements. For example, if one of your design requirements is that the total cost of the prototype cannot exceed X, then you could present a table of costs for each component of your proposed design solution as justification for why your design is likely to meet that requirement.

"Credible evidence" means it shouldn't just be your hunch that this will work. Be creative about how you validate the viability of your design. You should go beyond just having a stakeholder or expert reviewer say that the design seems viable. For example, you could point to evidence that a sub-system already functions, or you demo a proof of concept, or you've discussed with an expert that said, "I've seen something similar work in this other related situation...", etc.

Use as many rows as needed. Remember, this chart below may help you organize your ideas, but you will need to actually execute the "credible evidence."

Design Viability Description	Connection to Design Requirement	Credible Evidence?
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<i>Example:</i> The solution is inexpensive	<i>Example:</i> Design does not cost more than \$X	<i>Example:</i> Create a table of costs for each component of the solution.
This solution has enough height to reach the birdies	Design has at least 20 ft (6.096 m)	We are going to build it and then we will measure it out
The solution is easy to transport	Design is not heavy	We are going to build it and then we will weight it
This solution is resistant enough to stay intact with the impact of other objects	Design material is resistant	We are going to build it and then we will test the resistance through impact